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Tunnelling: Passing Through Walls

KKKC6353 Nanoelectronic Devices — Lecture 3, Document 1 of 3

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How to use this document

Lecture 2 showed that an electron behaves as a wave, and that a wave trapped in a box can have only certain energies. It ended with a hint: real walls are never perfect, and the wave leaks a little way into them. This lecture follows that leak all the way through the wall. The result is **tunnelling**, one of the strangest effects in physics and one of the most useful in electronics.

This document builds the picture:

- an electron can pass through a wall that, by everyday rules, it does not have the energy to climb;
- the chance of doing so depends very strongly on how **thick** and how **high** the wall is, and on how **light** the particle is;
- tunnelling powers the Sun, lets us see single atoms, leaks current out of transistors, and stores the photographs on your phone.

Document 2 uses tunnelling to make electronic devices: the tunnel diode, the resonant-tunnelling diode and the tunnel transistor. Document 3 shows what happens when electrons tunnel onto an island so small that adding even one of them matters.

Each section opens with a blue **Key ideas** box and closes with a few **check questions**. Green **Going further** boxes hold the mathematics behind the pictures. They are optional: read them if you want to see where the numbers come from, skip them if you do not. Boxes marked **A common misconception** point out ideas that sound right but are not.

> **From Lecture 2: what you need to remember**
>
> An electron is described by a wave, ψ. Where the wave is large, the electron is likely to be found; the chance is set by $|\psi|^2$.
>
> A wave trapped between walls can have only certain energies, and the smaller the space, the further apart they are.
>
> Real walls have a finite height. The wave does not stop dead at such a wall: it leaks a short way in and dies away (Lecture 2, Document 1, Section 6).
>
> If any of this is unfamiliar, read Lecture 2, Document 1 first.

1 A ball, a hill and an electron

> **Key ideas**
>
> A ball that does not have enough energy to get over a hill always rolls back.
>
> An electron meeting a wall it does not have the energy to climb sometimes appears on the other side.
>
> This is called tunnelling. It happens because the electron is a wave.

1.1 The everyday rule

Roll a ball towards a hill. If it is moving fast enough, it reaches the top and rolls down the far side. If not, it slows, stops part of the way up and rolls back. There is no third possibility. A ball with too little energy is *never* found on the far side of the hill, however many times you try.

Electrons meet “hills” all the time inside devices. They are not hills of earth but of energy: a thin layer of insulator between two metals, the junction between two semiconductors, or the gap of empty space between two surfaces. To cross such a layer the electron would need more energy than it has. Classical physics, the physics of the ball, says it must bounce back.

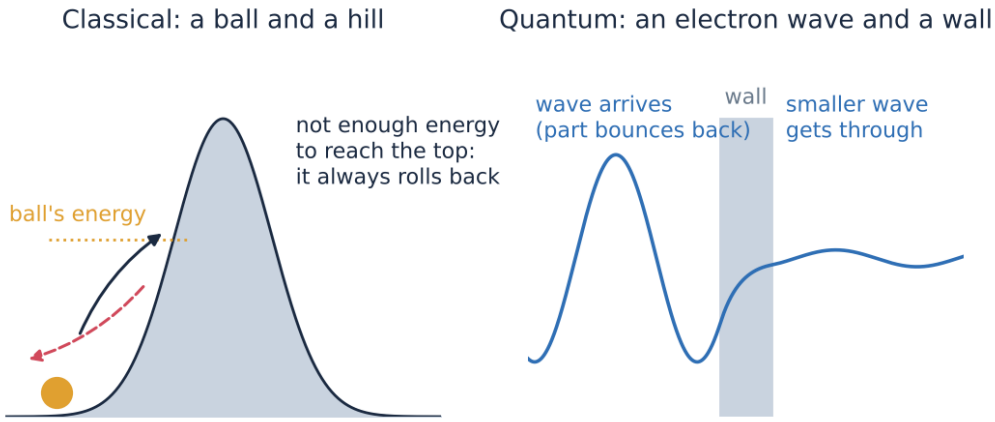


Figure 1.1. *Left: a ball without enough energy to reach the top always rolls back. Right: an electron wave meeting a wall it cannot climb. Most of the wave reflects, but a smaller wave appears on the far side. Figure made for this course.*

1.2 What electrons actually do

Electrons do not obey this rule. Send a beam of electrons at a thin enough wall and a few of them turn up on the far side, even though none of them had the energy to climb it. They have not gone over the top and they have not broken the wall. They have passed *through* it. This is **quantum tunnelling**.

The reason is the one Lecture 2 gave for almost everything strange about electrons: an electron is a wave. A ball is a small hard lump at one place; it either reaches the top or it does not. A wave is spread out, and when a wave meets a wall it does not simply stop. Part of it reflects, part of it enters the wall and dies away inside, and if the wall is thin enough, what is left of it comes out of the other side. Wherever there is some wave, there is some chance of finding the electron.

Light does the same thing. When light is reflected inside a glass block, a faint wave leaks a fraction of a wavelength beyond the surface. Bring a second block close enough, without touching, and some light jumps the gap. Optical engineers call this *frustrated total internal reflection*, and it is used in some fingerprint readers. Electrons do it because they are waves too.

Watch. A two-minute introduction from minutephysics: *What is quantum tunneling?*, youtube.com/watch?v=cTodS8hkSDg.

Check your understanding

1. A ball rolls towards a hill without enough energy to reach the top. What happens to it, and how is this different for an electron meeting a thin wall?
2. In one sentence, why can an electron do what the ball cannot?

2 What happens inside the wall

> **Key ideas**
> Inside the wall the electron's wave does not wiggle; it simply shrinks.
> What comes out has the same wavelength, and so the same energy, but it is smaller.
> The size of the wave that comes out sets the chance that the electron gets through.

2.1 Following the wave

The figure below follows an electron's wave, worked out exactly from the Schrödinger equation, as it meets a wall. Three things happen, one in each region.

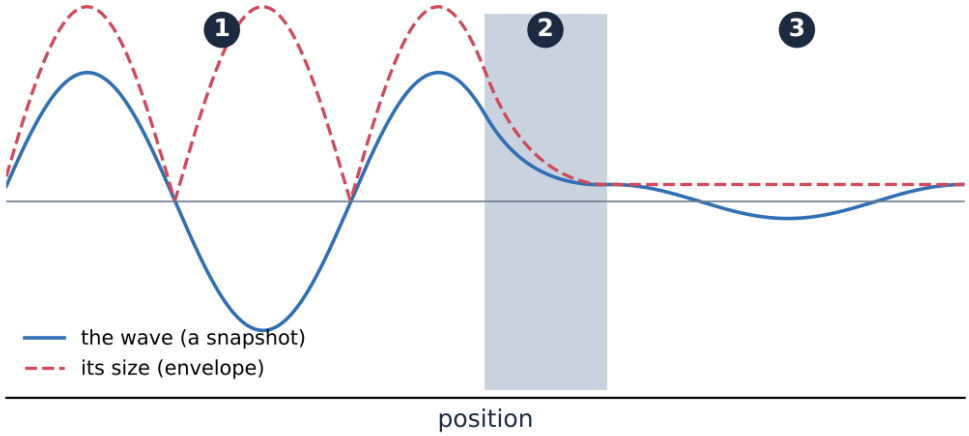


Figure 1.2. *An electron wave meeting a wall (shaded) that is higher than its energy. (1) On the left the incoming and reflected waves overlap. (2) Inside the wall the wave does not wiggle; its size dies away smoothly. (3) On the right a much smaller wave leaves with the same wavelength. The dashed line shows the size of the wave. Figure made for this course, from an exact solution.*

1. **Before the wall** the wave arrives and most of it bounces back. The arriving and returning waves overlap, which is why the dashed line goes up and down.
2. **Inside the wall** the electron does not have enough energy to "be" there in the ordinary sense, and the wave stops wiggling. Instead it fades away smoothly, a little like the sound of a bell dying out. The steeper the fade, the less of the wave reaches the far side.
3. **After the wall** whatever wave is left carries on as an ordinary wave again. It has exactly the same wavelength as before, which means the electron has exactly the same energy. It is just much smaller.

The chance of finding the electron on the far side is set by the size of that final wave, squared. This chance is called the **transmission probability**, written T. If T is 0.01, one electron in a hundred gets through and the other ninety-nine bounce back.

A common misconception

"The electron loses energy pushing through the wall." It does not. The wave that comes out has the same wavelength as the one that went in, so the electron has the same energy. Tunnelling is not a struggle through the wall that tires the electron out; it is all or nothing.

Either the electron is found on the far side, with all its energy, or it is found back where it started.

2.2 The wall must be thin

The fading inside the wall is fast. For an electron facing a wall 1 eV higher than its energy, the wave shrinks to about a third of its size for every 0.2 nm it travels into the wall. That is why tunnelling is a *nanoscale* effect. A wall a few nanometres thick lets a measurable number of electrons through; a wall a micrometre thick lets through essentially none, ever.

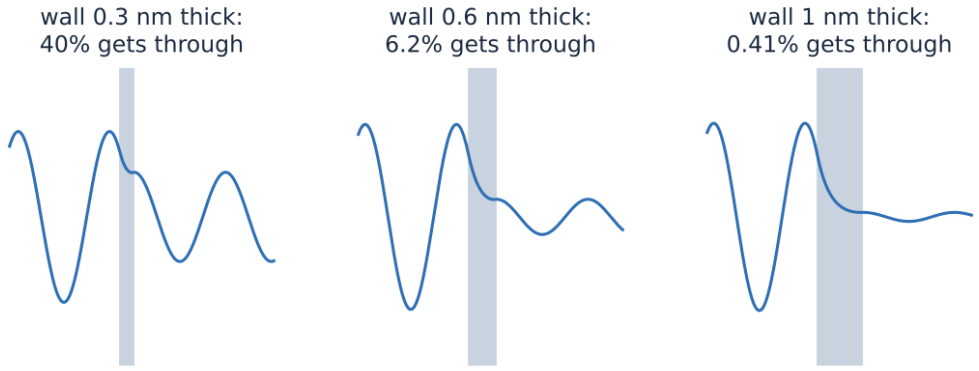


Figure 1.3. *The same electron meeting walls of three thicknesses. Doubling the thickness from 0.3 to 0.6 nm cuts the chance of getting through about six-fold; at 1 nm only about 4 in 1000 get through. Figure made for this course, from exact solutions.*

Going further: the fading rate (optional)

Inside a wall of height V_0, an electron of energy $E < V_0$ has a wave that fades as $\psi \propto e^{-\kappa x}$, with

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

For a free electron ($m = m_0$) and $V_0 - E = 1$ eV, $\kappa \approx 5.1$ nm^{-1}: the wave falls by a factor $e \approx 2.7$ every 0.2 nm.

For a wall of thickness d with κd larger than about 1, the transmission probability is approximately

$$T \approx 16 \, \frac{E(V_0 - E)}{V_0^2} \, e^{-2\kappa d}.$$

The factor in front is of order 1. Almost everything is in the exponential: T falls by $e^{-2\kappa}$ for every extra nanometre, which for a 1 eV wall is a factor of about 30 000. This is the most important formula in this lecture.

Check your understanding

1. Describe what the electron's wave does in each of the three regions: before, inside and after the wall.
2. An electron tunnels through a wall. Does it come out with more, less or the same energy? How do you know?
3. Why is tunnelling important in nanometre devices but never seen in a centimetre-sized one?

Explore. The animation *Tunnelling through a wall* (faizverse.github.io/nanoelectronic-devices/L3) lets you send a wave packet at a wall and change its thickness and height.

3 What controls the odds

> **Key ideas**
>
> Three things decide how easily a particle tunnels: the thickness of the wall, its height, and the mass of the particle.
>
> Thickness matters most. A wall twice as thick does not halve the chance; it can cut it ten-thousand-fold.
>
> Light particles tunnel easily. Electrons, the lightest particles in a device, tunnel far more readily than atoms.

3.1 Thickness

The chance of getting through does not fall in proportion to the thickness. It falls by the same *factor* for every extra slice of wall. Mathematicians call this an **exponential** fall. The table shows what that means for an electron facing a wall 1 eV higher than its energy.

Wall thickness	Chance of getting through
0.5 nm	about 1 in 50
1 nm	about 1 in 8 000
1.5 nm	about 1 in 1.3 million
2 nm	about 1 in 200 million
3 nm	about 1 in 6 million million

Going from 1 nm to 2 nm does not halve the current. It cuts it about 30 000 times. This extreme sensitivity is the key to almost every application in this lecture. It makes tunnelling devices very good at sensing tiny distances, and it makes thin insulators in ordinary transistors very hard to control.

3.2 Height and mass

A higher wall makes the wave fade faster inside it, so fewer electrons get through. For a 1 nm wall, raising its height above the electron from 0.5 eV to 3 eV cuts the chance from about 1 in 350 to about 1 in 26 million.

Mass matters in the same way. A heavier particle's wave fades much faster. A proton is about 1840 times heavier than an electron. Through a wall only 0.1 nm thick and 1 eV high, three out of four electrons get through, but only about one proton in seven million million. This is why we talk about electrons tunnelling in devices and almost never about atoms.

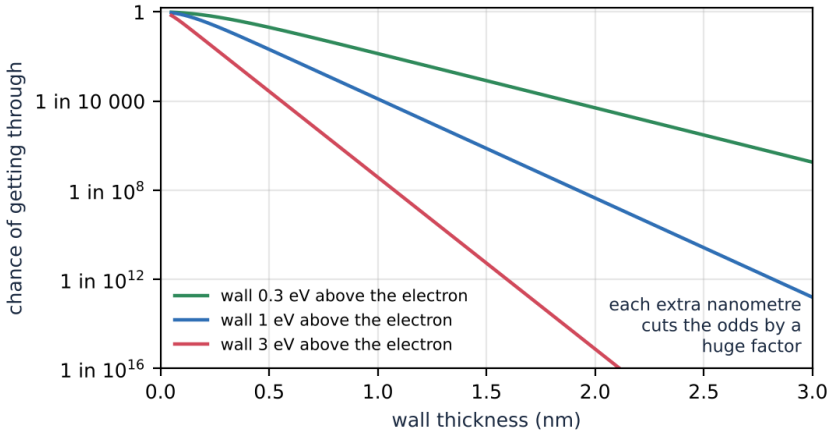


Figure 1.4. *The chance of tunnelling against wall thickness, for an electron meeting walls of three heights. The vertical scale is logarithmic: each step down is ten thousand times smaller. Thicker walls and higher walls both cut the chance enormously. Figure made for this course, from exact solutions.*

Inside a semiconductor, the electron's *effective mass* (Lecture 2, Document 1) is what counts. In GaAs it is only 0.067 times the free-electron mass, so electrons tunnel through much thicker layers there than in a vacuum gap of the same height. Device designers use this, as Document 2 shows.

> **Going further: three rules of thumb (optional)**
>
> From $T \approx e^{-2\kappa d}$ with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$:

- **Thickness** d sits in the exponent directly: doubling d squares the (small) transmission.
- **Height** enters as $\sqrt{V_0 - E}$: four times the height above the electron doubles κ.
- **Mass** enters as $\sqrt{m}$: a proton ($m \approx 1836\, m_0$) has a κ about 43 times larger than an electron's.

> A useful way to think about it: tunnelling is noticeable when κd is no more than a few, that is, when the wall is no thicker than a few "fade lengths" $1/\kappa$. For electrons and walls of about 1 eV, the fade length is about 0.2 nm.

Check your understanding

1. Name the three things that decide how easily a particle tunnels.
2. A tunnel barrier is made 0.5 nm thicker. Roughly what happens to the current: does it halve, or fall by a much larger factor? Use the table.
3. Why do electrons tunnel in devices while the atoms around them do not?

4 Tunnelling in nature

> **Key ideas**
>
> Tunnelling is not just a laboratory curiosity. The Sun shines because of it.
>
> Some radioactive atoms decay because particles tunnel out of their nuclei.
>
> Both were understood in the late 1920s, among the first great successes of quantum mechanics.

4.1 Why the Sun shines

The Sun is powered by **nuclear fusion**: in its core, hydrogen nuclei (protons) join together, and in a chain of steps make helium, releasing energy. To fuse, two protons must get extremely close. But both are positively charged, and they repel each other more and more strongly as they approach. This repulsion is a wall of energy.

The centre of the Sun is about 15 million degrees, and its protons move very fast. Even so, a typical proton has well under one per cent of the energy needed to climb that wall. By classical physics the Sun should not shine at all. Protons fuse because they **tunnel** through the last part of the wall. Each proton's chance is tiny, but the Sun contains so many protons, colliding so often, that the trickle of successful tunnelling events powers it steadily for billions of years. If tunnelling were any easier the Sun would burn out quickly; if it were any harder, it would be dark.

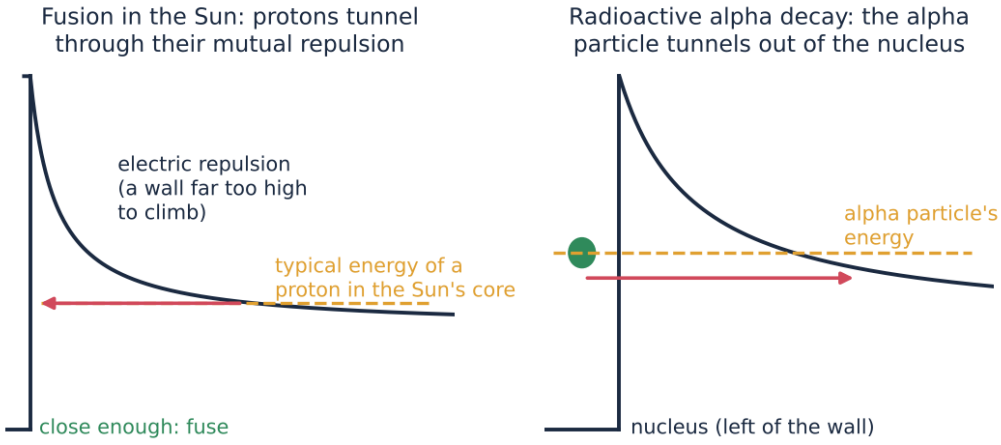


Figure 1.5. *Left: two protons repel each other; the wall is far higher than their typical energy, but they tunnel through its outer part and fuse. Right: an alpha particle trapped in a nucleus tunnels out through the same kind of wall. Figure made for this course; not to scale.*

4.2 Radioactive decay

The same idea, run backwards, explains a kind of radioactivity. Some heavy nuclei, such as uranium-238, spit out an **alpha particle** (two protons and two neutrons). Inside the nucleus the alpha particle is held in by the strong nuclear force; outside, it is pushed away by electric repulsion. Between the two there is a wall it does not have the energy to climb. In 1928 George Gamow, and independently Ronald Gurney and Edward Condon, showed that it tunnels out.

Their explanation solved a puzzle. The half-lives of alpha emitters range from less than a millionth of a second to billions of years, yet the energies of the alpha particles they emit

differ by only about a factor of two. The exponential sensitivity of tunnelling explains this enormous spread: a slightly more energetic alpha particle faces a thinner and lower part of the wall, and gets out vastly sooner.

Check your understanding

1. Why should the Sun not shine at all according to classical physics?
2. Alpha emitters with similar energies can have half-lives that differ by a factor of a million million. What property of tunnelling explains this?

5 Seeing atoms: the scanning tunnelling microscope

> **Key ideas**
>
> A scanning tunnelling microscope (STM) brings a sharp metal tip within about a nanometre of a surface and measures the tiny tunnelling current.
>
> Because tunnelling is so sensitive to distance, the current changes about ten times when the gap changes by one-tenth of a nanometre.
>
> Scanning the tip across the surface while keeping the current steady traces out the shape of individual atoms.

5.1 The idea

In 1981 Gerd Binnig and Heinrich Rohrer, at IBM's laboratory in Zurich, turned the extreme sensitivity of tunnelling into a microscope. They shared the 1986 Nobel Prize in Physics for it.

The **scanning tunnelling microscope**, or STM, has a metal tip so sharp that, ideally, it ends in a single atom. The tip is brought within about 0.5 to 1 nm of a conducting surface, without touching it. The gap is empty space, a wall for the electrons. A small voltage is applied, and a current of about a nanoampere flows as electrons **tunnel** across the gap.

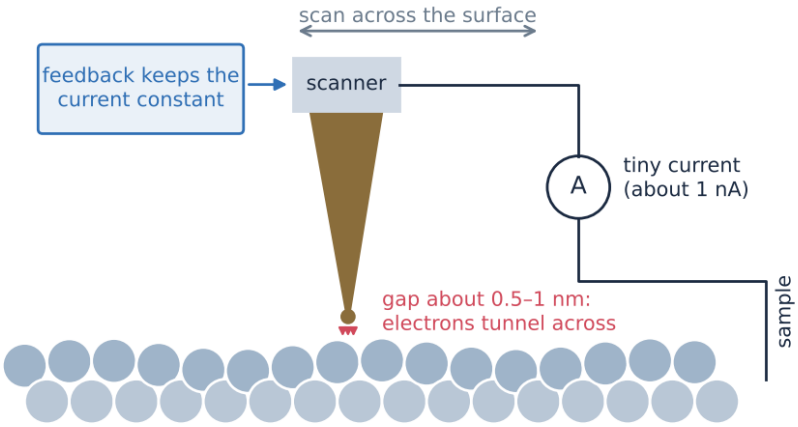


Figure 1.6. *A scanning tunnelling microscope. A sharp tip hovers about a nanometre above the surface. Electrons tunnel across the gap, and a feedback circuit raises or lowers the tip to keep the current constant as it scans. The tip's up-and-down movement traces the surface. Figure made for this course.*

The current depends on the size of the gap in exactly the exponential way of Section 3. Move the tip one-tenth of a nanometre closer, much less than the size of an atom, and the current

rises about ten times. So, as the tip scans across the surface, a feedback circuit moves it up and down to keep the current constant. When it passes over an atom it rises; between atoms it dips. Record the tip's height at every point, and you have a map of the surface with single atoms visible. The tip is moved by **piezoelectric** crystals, which change size very slightly when a voltage is applied, and can position it to a hundredth of a nanometre.

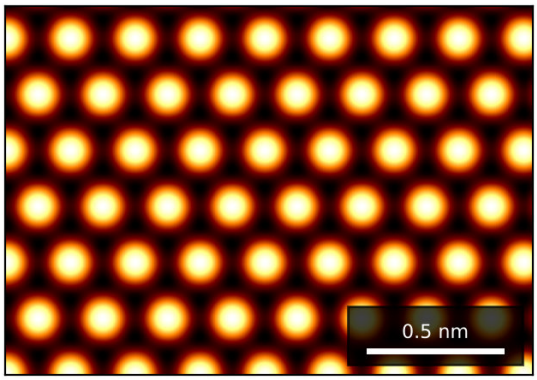


Figure 1.7. *What an STM image of a crystal surface looks like: each bright spot is an atom (here a simulated hexagonal surface with the spacing of graphite, about 0.25 nm). Figure made for this course; simulated, not measured.*

The STM opened the field of nanotechnology. With it, researchers have pushed single atoms around a surface to build structures one atom at a time, and measured the energy levels of individual molecules. It works only on conducting surfaces; its close cousin, the **atomic force microscope**, feels the force on the tip instead and works on insulators too.

Explore. The animation *Scanning tunnelling microscope* (faizverse.github.io/nanoelectronic-devices/L3) lets you scan a tip over a row of atoms and watch the current and the image.

> **Going further: why the STM is so sensitive (optional)**
>
> The tunnelling current is roughly $I \propto V\, e^{-2\kappa d}$, where d is the gap. For a vacuum gap the wall height is set by the *work function* of the metal, about 4 to 5 eV. With 4.5 eV, $\kappa \approx 10.9\ \text{nm}^{-1}$, so a change of 0.1 nm in d changes the current by $e^{2\times 10.9\times 0.1} \approx 9$.
>
> An STM electronics chain can easily hold the current steady to within a few per cent, which corresponds to keeping the gap steady to about one-thousandth of a nanometre. That is why it sees atoms, which are about 0.2 to 0.3 nm across, so clearly.

Check your understanding

1. What crosses the gap between the STM tip and the surface, and why does anything cross at all?
2. Why does the STM keep the current constant and move the tip, rather than holding the tip still and measuring the current?

6 Tunnelling as a problem: leaky transistors

> **Key ideas**
>
> A transistor's gate is separated from the channel by a very thin insulator.
>
> When that insulator became only about 1.2 nm thick, electrons started tunnelling straight through it, wasting power.
>
> The solution was a new insulator, physically thicker but just as effective electrically.

6.1 The thinning gate insulator

Lecture 1 described how transistors were shrunk generation after generation. One part that shrank was the insulating layer between the gate and the silicon channel, made of silicon dioxide (SiO_2). A thinner layer lets the gate control the channel more strongly, which is what scaling needs.

By the early 2000s this layer was about 1.2 nm thick, only four or five atoms. Such a layer is a wall of the kind this document has been describing, and electrons began to tunnel straight through it from the gate to the channel. Each transistor leaked only a tiny current, but a chip has billions of them, and the leakage became a serious drain on power.

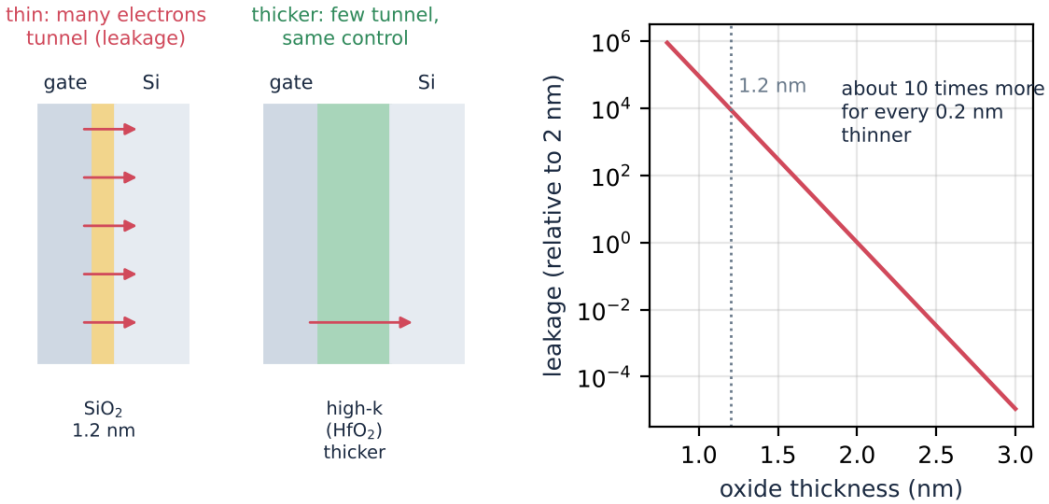


Figure 1.8. *Left: a very thin SiO_2 gate insulator lets many electrons tunnel through; a thicker high-k insulator gives the same control with far less leakage. Right: in a simple model, tunnelling leakage grows about ten times for every 0.2 nm the oxide is thinned. Figure made for this course; the right panel uses an electron barrier of 3.1 eV and an effective mass of 0.4 m_0.*

The exponential rule made this impossible to fix by small adjustments. Thinning the oxide from 2 nm to 1.2 nm raises the tunnelling current by roughly ten thousand times. Thinning it any further was out of the question.

6.2 The fix: high-k

The answer, introduced by Intel at the 45 nm generation in 2007 and described in Lecture 1, Document 1, was to replace SiO_2 with a hafnium-based material whose *dielectric constant*, k, is five to six times higher. A higher k means a thicker layer gives the gate the same grip on the channel. So the insulator could be made several times thicker, and because tunnelling

falls exponentially with thickness, the leakage fell by more than a hundred times. Every advanced chip made today uses such a “high-k” gate insulator.

Check your understanding

1. Why did electrons start to leak through the gate insulator of transistors in the early 2000s?
2. How does using a high-k insulator reduce the leakage without weakening the gate’s control?

7 Tunnelling as a tool: memory

> **Key ideas**
>
> Flash memory stores data as electrons trapped on an isolated layer, put there and removed by tunnelling.
>
> A strong electric field tilts the wall into a thin triangle, so electrons can tunnel through a thick insulator only when we want them to.
>
> Magnetic memory and hard-disk read heads use a different kind of tunnelling, through a 1 nm layer between two magnets.

7.1 Flash memory

The photographs on your phone and the files on a solid-state drive are stored in **flash memory**. Each memory cell is a transistor with an extra layer: a *floating gate* (or, in most modern chips, a charge-trapping layer) completely surrounded by insulator. Electrons placed on it stay there for years, because the insulator around them, about 8 nm thick, is far too thick for them to tunnel out. The trapped charge changes how easily the transistor turns on, and that is read as a 0 or a 1.

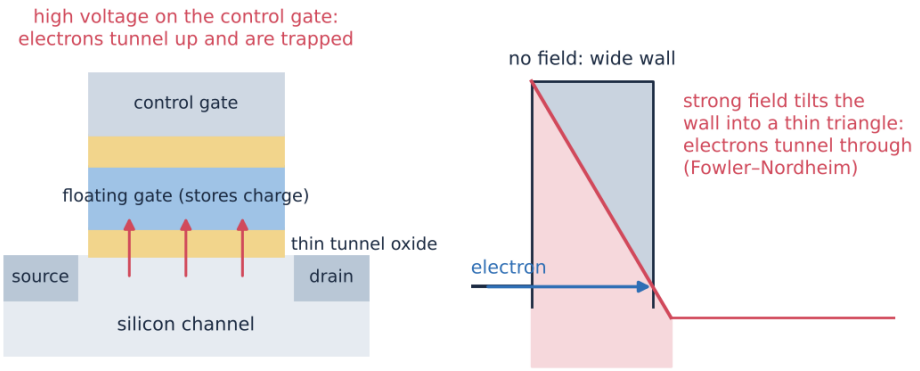


Figure 1.9. *Left: a flash memory cell. Electrons are pushed onto the floating gate through a thin tunnel oxide by a high voltage on the control gate. Right: the strong field tilts the energy wall into a thin triangle, through which electrons tunnel. Without the field, the wall is too thick and the stored electrons stay put for years. Figure made for this course.*

How do electrons get onto a layer that is completely sealed in? By tunnelling, under a trick. A high voltage, around 20 V, is put on the control gate. The strong electric field tilts the energy wall, so that instead of a thick rectangle the electrons face a thin triangle. They tunnel through the thin tip of the triangle onto the floating gate. This is called **Fowler–Nordheim tunnelling**, after the two physicists who explained it in 1928. Reverse the

voltage, and the electrons tunnel back out, erasing the cell. With no voltage, the wall is thick again, and the data stays put.

Each write and erase pushes electrons through the oxide and slowly damages it. This is why flash memory wears out after many thousands of rewrites, and why solid-state drives spread their writes across all their cells.

7.2 Magnetic tunnel junctions

A **magnetic tunnel junction** is a sandwich of two magnetic layers separated by an insulator only about 1 nm thick, usually magnesium oxide. Electrons tunnel through the insulator, but how easily depends on whether the two magnets point the same way or opposite ways. The resistance can differ by a factor of two or more between the two cases.

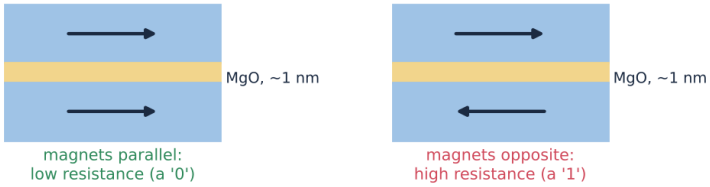


Figure 1.10. *A magnetic tunnel junction. When the two magnetic layers point the same way, electrons tunnel easily (low resistance); when they point opposite ways, much less easily (high resistance). Figure made for this course.*

This effect has been used since the mid-2000s in the read heads of hard disks, which sense the tiny magnetic fields of the recorded bits. It is also the basis of **MRAM**, a memory that keeps its data without power, now built into some microcontrollers and chips for cars and industry.

Check your understanding

1. How do electrons get onto a floating gate that is completely surrounded by insulator?
2. Why do the stored electrons not leak back out when the power is off?
3. What decides the resistance of a magnetic tunnel junction?

8 Summary, and what comes next

1. **An electron can pass through a wall it cannot climb**, because it is a wave. Part of the wave fades through the wall and emerges on the far side.
2. **The electron keeps all its energy.** Tunnelling is all or nothing.
3. **Thickness matters most.** The chance of tunnelling falls exponentially: each extra nanometre can cut it tens of thousands of times. Higher walls and heavier particles also tunnel far less.
4. **Nature uses tunnelling.** It lets the Sun shine and explains alpha decay.
5. **The STM** turns the exponential sensitivity into a microscope that sees single atoms.
6. **In transistors, tunnelling is a problem**: it leaks current through thin gate insulators, which is why chips switched to high-k materials.

7. **In memory, tunnelling is a tool**: flash memory writes and erases by Fowler–Nordheim tunnelling, and magnetic tunnel junctions read hard disks and store data in MRAM.

Where Document 2 begins. If the tunnelling current can be switched on and off by a voltage, it can be used to build devices. Document 2 shows how heavily doped p–n junctions, layered semiconductors and a new kind of transistor do exactly that, and how one of them produces a current that *falls* as the voltage rises.

Words to know

Word	Plain meaning
Tunnelling	A particle passing through an energy wall it does not have the energy to climb
Barrier	The "wall": a region where the electron's energy is lower than the potential energy
Transmission probability, T	The chance that an electron arriving at a barrier gets through
Exponential fall	Falling by the same factor for each equal step
STM	Scanning tunnelling microscope: images atoms using the tunnelling current to a sharp tip
High-k dielectric	A gate insulator that can be made thicker than SiO_2 for the same effect, so it leaks less
Fowler–Nordheim tunnelling	Tunnelling through the thin tip of a barrier tilted by a strong electric field
Floating gate	A sealed conducting layer in a flash cell that stores electrons
Magnetic tunnel junction	Two magnets separated by a 1 nm insulator; its resistance depends on their directions

Sources and further reading

Gentle introductions

- minutephysics, *What is quantum tunneling?*, youtube.com/watch?v=cTodS8hkSDg.
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Textbooks, for the Going further boxes

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Figures. All figures were made for KKKC6353 (scripts in `figures-source/`). The numbers in the tables and graphs are exact solutions of the Schrödinger equation for rectangular barriers.